Software Requirement Specifications

<Project Title>

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Summary

This document outlines the detailed functional and non-functional requirements for the proposed Network Monitoring System (NMS). The NMS is designed to provide comprehensive, real-time supervision of all connected network devices and resources within an organization. It aims to proactively detect faults, analyze performance bottlenecks, provide automated alerts, and offer centralized control and reporting to the network administration team, thereby ensuring high network availability and performance.

Table of Contents

1. Introduction

A Network Monitoring System (NMS) is a critical framework used by organizations to track the **health, performance, and security status** of their entire IT infrastructure. It ensures the effective operation of a network and is a central part of network management.

1.1. Purpose

The primary purpose of this SRS is to clearly define the requirements for developing a Network Monitoring System. The NMS's main goal is to enable **proactive maintenance** by instantly identifying problems like overloaded servers, high error rates on routers, or unusual traffic spikes, allowing IT personnel to address them before they impact service.

1.2. Scope

The NMS will supervise and manage all connected computers, servers, routers, switches, and firewalls within the network, whether on-premises or in a data center. It will cover:

- **Fault Detection** and **Performance Analysis**.
- **Real-time** performance metrics collection and visualization.
- **Alerting** administrators to performance degradation or device outages.

1.3. Product Perspective

The NMS is a **client-server-based project**. It consists of two main components:

- **Management Station (Server):** This is the central control point where network administrators monitor, analyze data, and generate reports.
- **Managed Devices (Clients):** These are the network elements (e.g., routers, servers, printers) being monitored. Each device must host an **SNMP Agent** or similar client program responsible for collecting and transmitting data.

1.4. User Characteristics

1.5. User Role	**1.6. Description**	**1.7. Key Activities**
1.8. Network Administrator	**1.9.** The primary user, with advanced technical knowledge of network protocols and devices.	**1.10.** Configure monitoring thresholds, analyze performance reports, manage alerts, perform remote troubleshooting.
1.11. IT Manager	**1.12.** Management user, focused on high-level availability, compliance, and historical trends.	**1.13.** View summary dashboards, review historical performance data, and check compliance reports.

1.14. Similar apps and systems/Literature Review

Modern network monitoring solutions employ various techniques for data collection:

SNMP-based Tools: Use the Simple Network Management Protocol (SNMP) to track real-time resource utilization (CPU, memory, bandwidth).

Flow-based Tools: Monitor traffic flow (e.g., NetFlow, sFlow) to provide statistics on protocols, users, and bandwidth consumption, aiding in bottleneck identification.

Active Monitoring: Solutions that inject test packets into the network to measure end-to-end performance metrics like latency, packet loss, and link utilization from a user's perspective.

1.15. Proposed Technologies

The system will primarily utilize the following industry-standard protocols and techniques:

Simple Network Management Protocol (SNMP) (v3 for security).

Internet Control Message Protocol (ICMP) (for basic availability checks like `ping`).

Secure Shell (SSH) (for secure configuration and command execution on Unix/Linux servers).

Database: For storing historical monitoring data and logs.

2. Requirements

2.1. Function Requirements

ID	Name	Description
FR001	Device Discovery & Mapping	The system shall automatically discover new devices on the network and allow the administrator to classify and logically map them.
FR002	Real-Time Performance Monitoring	The system shall continuously collect and display key metrics in real-time, including CPU usage, Memory consumption, Bandwidth utilization, latency, and packet loss for all managed devices.
FR003	Availability Monitoring	The system shall check the uptime (availability) of all network devices and services (e.g., web, email) at defined intervals.

ID	Name	Description
FR004	Intelligent Alerting	The system shall allow the administrator to set customizable thresholds for any monitored metric and automatically trigger an alert (via email or SMS) when a threshold is violated.
FR005	Centralized Dashboard	The system shall provide a single-pane-of-glass dashboard to visualize the network's overall status, active alerts, and real-time performance graphs.
FR006	Historical Reporting	The system shall store monitoring data for at least one year and generate customizable historical reports on performance, availability, and traffic flow for trend analysis and capacity planning.
FR007	Remote Management	The system shall provide remote capabilities to the administrator, such as initiating a remote shutdown, reboot, or logoff on managed workstations.

2.2 Non-Functional Requirements

ID	Category	Requirement
NFR001	Performance (Response Time)	The monitoring dashboard shall update with real-time data within 5 seconds of the data being collected from the agent.
NFR002	Performance (Throughput)	The Management Station shall be capable of handling at least 10,000 requests (SNMP GET/Trap/SET) per minute without performance degradation.
NFR003	Reliability / Availability	The system shall maintain a minimum service availability (uptime) of 99.9%. The Management Station must support a redundant configuration for continuous monitoring.
NFR004	Scalability	The system architecture shall be designed to scale and support a 25% growth in the number of monitored devices without requiring a full system redesign.
NFR005	Security	All network communication between the agent and management station for configuration and control (e.g., remote reboot) shall be secured using encryption protocols (e.g., SSH or SNMPv3).
NFR006	Usability	The administrator interface (dashboard and configuration panels) must be intuitive and allow a trained administrator to configure a new monitoring agent within 10 minutes.

3. Use Cases and Flow of Processes

Use cases are the formal representation of the process flow defined by functional requirements. The diagram below illustrates the high-level functions of the Network Monitoring System.

Figure 1: System Level Use Case Diagram

3.1. Use Case: Monitor Device Performance

Field	Value
ID	UC001
Name	Monitor Device Performance
Description	This use case describes how the administrator uses the system's dashboard to check the real-time status and performance of a specific network device.
Requirement(s)	FR002, FR004, FR005
Actor(s)	Network Administrator
Precondition	The device is online, has the NMS agent running, and is actively being monitored. The Administrator is logged into the Management Station.
Postcondition	The Administrator views the current performance metrics of the device and identifies its health status.
Basic Flow	1. Administrator selects the desired device from the network map or device list on the centralized dashboard. 2. System sends a request (e.g., SNMP GET) to the device's agent to retrieve current performance metrics (CPU, Memory, Traffic).

Field	Value
	3. Device Agent responds with the requested data. 4. System immediately displays the metrics in real-time graphs and health status indicators on the administrator's screen.
Alternative Flow: Alert Generated	1. During data collection (Basic Flow Step 3), the System's monitoring logic detects that the CPU usage has exceeded the pre-set threshold (e.g., >80%). 2. System generates a Critical Alert (FR004). 3. System immediately sends an email notification to the Administrator. 4. The alert is displayed on the dashboard, prompting the Administrator to investigate.
Exceptions	Device Offline: The device does not respond to the system's request, and the system flags a "Device Down" fault alarm.

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